KAIZEN MATHS

Lesson starter

Transposition of Formulae: Changing the Subject

Subject Repeated

Name _____

S1

Collect the terms containing the subject, then make that variable the subject.

Make x the subject.

$y = 6x + 5 - 2x$

S2

Move all terms containing the subject to one side, then factorise.

Make x the subject.

$y + 6x = 19 - 4x$

S3

Remove the denominator first, then isolate the subject.

Make x the subject.

$y = \frac{12}{x+2}$

S4

Remove the denominator, collect the subject terms, factorise, then divide.

Make x the subject.

$y = \frac{2x+4}{x+3}$

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ALG-0852-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Subject Repeated

Name _____

EA1

Collect the terms containing the subject, then make that variable the subject.

Make x the subject.

$y = 9x + 3 - 2x$

EA2

Move all terms containing the subject to one side, then factorise.

Make x the subject.

$y + 5x = 12 - 7x$

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Subject Repeated

Name _____

EB1

Remove the denominator first, then isolate the subject.

Make x the subject.

$y = \frac{10}{x+7}$

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Subject Repeated

Name _____

EA1

Collect the terms containing the subject, then make that variable the subject.

Make x the subject.

$y = 9x + 3 - 2x$

EA2

Move all terms containing the subject to one side, then factorise.

Make x the subject.

$y + 5x = 12 - 7x$

ALG-0852-L3 · EA

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Subject Repeated

Name _____

EB1

Remove the denominator first, then isolate the subject.

Make x the subject.

$y = \frac{10}{x+7}$

ALG-0852-L3 · EB

Show your working

Teacher answers & working

Transposition of Formulae: Changing the Subject · Subject Repeated

ALG-0852-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Collect the terms containing the subject, then make that variable the subject.

Make x the subject.

$y = 6x + 5 - 2x$

Answer:

$x = \frac{y-5}{4}$

Working:

Start with the formula and identify all terms containing x.

$y = 6x + 5 - 2x$

Collect the x terms: 6x - 2x = 4x.

$y = 4x + 5$

Subtract 5 from both sides.

$y - 5 = 4x$

Divide both sides by 4.

$\frac{y-5}{4} = x$

Write x as the subject.

$x = \frac{y-5}{4}$

S2

Move all terms containing the subject to one side, then factorise.

Make x the subject.

$y + 6x = 19 - 4x$

Answer:

$x = \frac{19-y}{10}$

Working:

Start with the formula. The subject x appears on both sides.

$y + 6x = 19 - 4x$

Add 4x to both sides so all x terms are on the left.

$y + 6x + 4x = 19$

Collect the x terms on the left.

$y + 10x = 19$

Subtract y from both sides.

$10x = 19 - y$

Divide both sides by 10.

$x = \frac{19-y}{10}$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Subject Repeated

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S3

Remove the denominator first, then isolate the subject.

Make x the subject.

$y = \frac{12}{x+2}$

Answer:

$x = \frac{12}{y} - 2$

Working:

Start with the formula. The subject is in the denominator, so remove the denominator first.

$y = \frac{12}{x+2}$

Multiply both sides by x + 2.

$y(x + 2) = 12$

Divide both sides by y to isolate the bracket.

$x + 2 = \frac{12}{y}$

Subtract 2 from both sides.

$x = \frac{12}{y} - 2$

S4

Remove the denominator, collect the subject terms, factorise, then divide.

Make x the subject.

$y = \frac{2x+4}{x+3}$

Answer:

$x = \frac{4-3y}{y-2}$

Working:

Start with the formula. The subject appears in the numerator and denominator.

$y = \frac{2x+4}{x+3}$

Multiply both sides by x + 3 to remove the denominator.

$y(x + 3) = 2x + 4$

Expand the left-hand side.

$xy + 3y = 2x + 4$

Move the x terms to one side and the non-x terms to the other side.

$xy - 2x = 4 - 3y$

Factorise the left-hand side by taking out x.

$x(y - 2) = 4 - 3y$

Divide both sides by y - 2.

$x = \frac{4-3y}{y-2}$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Subject Repeated

ALG-0852-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

Collect the terms containing the subject, then make that variable the subject.

Make x the subject.

$y = 9x + 3 - 2x$

Answer:

$x = \frac{y-3}{7}$

Working:

Start with the formula and identify all terms containing x.

$y = 9x + 3 - 2x$

Collect the x terms: 9x - 2x = 7x.

$y = 7x + 3$

Subtract 3 from both sides.

$y - 3 = 7x$

Divide both sides by 7.

$\frac{y-3}{7} = x$

Write x as the subject.

$x = \frac{y-3}{7}$

EA2

Move all terms containing the subject to one side, then factorise.

Make x the subject.

$y + 5x = 12 - 7x$

Answer:

$x = \frac{12-y}{12}$

Working:

Start with the formula. The subject x appears on both sides.

$y + 5x = 12 - 7x$

Add 7x to both sides so all x terms are on the left.

$y + 5x + 7x = 12$

Collect the x terms on the left.

$y + 12x = 12$

Subtract y from both sides.

$12x = 12 - y$

Divide both sides by 12.

$x = \frac{12-y}{12}$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Subject Repeated

ALG-0852-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Remove the denominator first, then isolate the subject.

Make x the subject.

$y = \frac{10}{x+7}$

Answer:

$x = \frac{10}{y} - 7$

Working:

Start with the formula. The subject is in the denominator, so remove the denominator first.

$y = \frac{10}{x+7}$

Multiply both sides by x + 7.

$y(x + 7) = 10$

Divide both sides by y to isolate the bracket.

$x + 7 = \frac{10}{y}$

Subtract 7 from both sides.

$x = \frac{10}{y} - 7$